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Sabbir Mahmud

Place of birth: Sylhet, Bangladesh **Phone:** (+880) 1629875968 (Mobile) **Email address:** mahmudsabbir222@gmail.com **LinkedIn:** Sabbir Mahmud **Address:** North Badda, Gulshan , 1212, Dhaka , Bangladesh (Home)

ABOUT ME

Applied Chemistry and Chemical Engineering graduate with industrial training and research experience. Interested in materials science, process engineering, energy systems, and sustainable technologies, with a focus on practical industrial applications and sustainable development.

WORK EXPERIENCE

Training Institute of Chemical Industries (TICI) - Bangladesh, Narsingdi

Website: https://tici.gov.bd/

Trainee- Chemical Engineering

[03/01/2026 - 29/01/2026]

- Completed four weeks industrial training (with A+ grade) on Industrial unit operation and process control techniques under Operation and Process Technology Department (OPTD)
- Gained practical exposure to process engineering and plant operations
- Developed skills in P&ID, ASPEN PLUS, Cascade control & PLC, SCADA control systems
- Learned industrial safety, hazard identification and fire protection systems
- Participated in workshops on water treatment, absorption, neutralization, Ion-exchange, heat exchangers, analytical, mechanical and electrical systems

American Chemical Society: Student Chapter DU - Bangladesh, Dhaka

Operating committee: Creative affairs

[01/03/2023 - 30/11/2024]

- Contributed to design, social media and communication materials for academic events
- Supported organizing scientific seminars, webinars and student workshop activities

EDUCATION & TRAINING

BSc. in Applied Chemistry and Chemical Engineering

University of Dhaka [10/01/2022 - 12/08/2026]

City: Dhaka | **Country:** Bangladesh

Website: https://www.du.ac.bd/body/ACT | **Final grade:** 3.5/4.0 | **Level in EQF:** 6

Higher Secondary Education Certificate (HSC)

Uttara Govt. College [01/01/2018 - 01/01/2020]

City: Dhaka | **Country:** Bangladesh

Website: https://www.ugc.edu.bd/ | **Field(s) of study:** Science: | **Final grade:** 5.0/5.0 | **Level in EQF:** 4

SKILLS

Aspen Plus, Aspen HYSYS | Documentation | Process optimization | Economic analysis | Data Analysis | Python | C/C++ | Communication, Collaboration

PUBLICATIONS

[2026]

Chitosan Based Hydrogels for Cr(VI) removal from waste water-A review

Reviews recent advances in chitosan-based hydrogels for Cr(VI) removal from wastewater. Examines adsorption mechanisms, material modification strategies, and regeneration performance for sustainable water treatment.

Authors: S.H. Nirob, S. Mahmud, J. Ali, T.U. Rashid | **Journal Name:** Materials Today Communications | **Volume, Issue and Pages:** Article: 116074, Vol 56 | **Publisher:** Elsevier

[2024]

Novel materials for hydrogen generation in contaminated water

Reviews emerging materials for hydrogen generation from contaminated water, focusing on catalytic mechanisms, pollutant removal, material stability, and regeneration for sustainable hydrogen production and wastewater treatment.

Authors: S.H. Nirob, S. Mahmud, N. Sakib, T.U. Rashid | **Journal Name:** Accelerating the Transition to a Hydrogen Economy | **Publisher:** Elsevier

Book: Accelerating the Transition to a Hydrogen Economy; Volume 1: Achieving Carbon Neutrality

PROJECTS

[01/08/2025 - 30/11/2025]

Commercialization of Perovskite-Based Solar Systems in Bangladesh: A Circular Economical Approach This project investigates Roll-to-Roll manufacturing as a scalable and cost-effective pathway for commercializing perovskite solar cells with a focus on Bangladesh's energy landscape. It covers perovskite material advantages, process control, regulatory and engineering considerations, estimation of economic feasibility, and the potential of perovskite-silicon tandem architectures.

[04/02/2026 - 17/05/2026]

Recovery of Cu/ZnO/Al_2O_3 Catalyst from Waste Printed Circuit Boards and Its Application in Industrial Methanol Synthesis Simulated an integrated Aspen Plus process on the recovery of Cu/ZnO/Al_2O_3 catalyst precursors from waste PCBs and their use in syngas-to-methanol synthesis. The study identifies the sub-stoichiometric H_2/CO ratio as the main process bottleneck and proposes water-gas shift integration and recycle loops for future improvement.

HONOURS AND AWARDS

[01/01/2022] DUMEA

Admission Excellence Award Awarded for outstanding academic performance leading to admission at University of Dhaka.

[01/01/2019] One Bank Limited

Scholarship Recipient Awarded merit-based educational scholarship for academic achievement.

VOLUNTEERING

Bangladesh National Chemistry Olympiad 2024

- Designed banners, standees, social media posts, and booklet covers for national chemistry competition
- Contributed visual materials supporting event communication and outreach